

3 dummy columns

color_Y	color_B	color_R	sum
1	0	0	1
0	1	0	1
0	0	1	1

drop

always $Y + B + R = 1$, so $Y = 1 - B - R$:
one column can be worked out from the others

$n - 1 = 2$ columns

color_B	color_R	
0	0	→ Yellow
1	0	→ Blue
0	1	→ Red

0, 0 still means Yellow:
no information is lost